

COMP5241 Project Overview

Each team may choose one of the following topics to complete the project.

Topic 1

With the popularity of generative AI models and APIs, many applications are integrating Generative AI capabilities into their applications (including the [GPTutor](#) platform that we have developed).

Here are some examples:

- [This AI Tool Does Things No One's Done Before | Napkin AI](#)
- [Introduction to Atlassian Intelligence in Jira | Atlassian](#)
- [Take teamwork to new heights with Mural AI + Microsoft Copilot](#)
- [Gamma: AI for presenting ideas, beautifully](#)
- [Introducing Notion AI](#)
- [How Quizizz AI works in 90 seconds](#)
- [MagicSchool - AI Built for Schools](#)

With the wide availability of GenAI APIs, hundreds of new GenAI powered apps are developed every week/month (e.g. Refer to the list of applications at <https://theresanaiforthat.com/tasks/>). In the market, there are many "GPT wrappers" where the apps simply provide a "superficial" layer of UI/UX functionality on top of GPT API but provide little additional values. Refers to the following webpages for some discussion on the issue.

- https://learnprompting.org/blog/2024/2/4/gpt_wrappers
- https://www.reddit.com/r/ChatGPT/comments/18mch6l/its_a_gpt_wrapper_have_no_value/
- https://www.reddit.com/r/SaaS/comments/17tt22f/why_do_people_keep_saying_open_ai_is_killing/

In this project, each group should select one focus area that your team is most interested in. You should critically evaluate the GenAI features in the existing apps. Identify the GenAI features which really delights the users and bring good values to the software product, versus those which are simply "GPT wrappers" with little values. Based on your study, identify the gap and develop a prototype which aims to enhance the existing applications and/or better leverage the power of "Generative AI" to enhance the software product and bring values to the users.

Topic 2

In the Department of Computing at PolyU, there are many courses that require students to complete a group project. One challenge that is faced by the instructor is evaluating the performance of the students in the different groups and the collaboration among the group members.

This project aims to develop a prototype of a web application that can help the instructor to track the progress of different groups in the course and the collaboration among the group members. By using the [GitHub Classroom](#), teachers should first create an "assignment," and the team leader of each group should accept the assignment to create a project repository for the team in the course project's GitHub Organization. The team members should add the other team members to the repository. The application will analyze various student activities in GitHub repositories, such as commits, issues, and pull requests. Dashboards and statistics will be provided to the instructor to provide insights about the activities of different groups and individual students' contributions in the project over time.

Here are some initial ideas about the project.

- Instructor may track the progress of different teams by summarizing the different activities in the repositories of the different groups, e.g. commit frequency, project board, issues, code changes, discussion/comments, assigned issues, pull requests, milestones, bug reporting, tasks completion, etc.
- The system should track the performance and activities and contribution of different students (e.g. identifying deadline fighters/free-riders). The system should be able to provide dashboard and generate reports for the instructor to track the progress of different groups and students' participation in the project.
- The system is expected to integrate with GitHub API to collect data from GitHub organization and repositories. You should investigate what data can be collected from GitHub API about students' activities and collaboration in the repository and how to present them to instructors.
- Generative AI may be used by the teacher to interact with the collected data and analyze the group/student's performance. You may experiment with the prompts that can be used for the different GenAI related features.

Just like real-world software projects, the initial requirements will often be vague initially and may change during the development process. The project is also open to any new ideas about the features of the application.

Group Formation

Students should form a group of 4-5 members in blackboard by the end of week 3 (i.e. 21/9/2024 (SAT)). Students without a group will be randomly assigned a group after the deadline. Each group should submit your topics (one per group) shortly after the group formation deadline.

Project GitHub Repository

The GitHub classroom invitation link will be available soon. Team leader is required to accept the invitation and create the project repository for the team in our course project's GitHub organization. The repository should be private and only accessible by the team members and the instructor. The team leader will be the admin of the repository and should the other team members to the repository for collaboration

During the project, you should use the GitHub repository (e.g. issues, project board, etc) to manage and collaborate on your project. Your project source code and other related files (e.g. documentation) should be uploaded to your project repository in the course project's GitHub organization.

Project discussion meeting

The project meeting will be held during the lectures in week 8 or week 9. Each group will have around 5 minutes to discuss your project progress. You should prepare a 1-2 page summary of your progress, division of work, and your preliminary system design/prototype for discussion. We may also clarify the requirements in the meeting.

Presentation

The presentation will be held in the week 12 and week 13. Each group will have around 8-10 minutes (to be confirmed) to present your project. The presentation slides should cover the following aspects:

- Review of related applications, their strength/limitations/pain points
- The demo and features of your prototype/system (record a video of your demo prototype)
- Reflection on the project
 - Discuss your development process and examples of how your group collaborates
 - Demonstrate with examples how you use GenAI to assist in your project development and implementation. Elaborate on what works well/not so well.
 - Reflect on what you have learnt/the findings from the project, what you have done well/not done so well, what the challenges are, etc.
- Appendix: You may include other information/screenshots that are relevant to your project.

Submission

- Presentation slide (with your code repository information) and zipped version of your GitHub repository (including all the source code, documentation, etc) should be submitted to the blackboard by the end of Week 11.

Evaluation Rubrics

- Project discussion meeting (5%)
- Final presentation and deliverables
 - System design and prototype (40%)
 - Presentation (40%)
 - ◐ Collaboration, sharing and reflection (15%): Workload distribution and collaboration using Github, Use of GenAI in the project, reflection (e.g. challenges, findings, reflection on the project), etc

FAQ

1. What programming languages should we use?

Ans: In the project, you are welcome to use any programming languages, frameworks, and libraries, etc. You can also GenAI for your project (e.g. idea brainstorming, designing, evaluating alternatives, development and coding, testing, etc).

2. Can I work on the project individually?

Ans: According to the COMP5241 subject description [form](#), the course's Intended Learning Outcome (ILO) d is about "Work and collaborate effectively in teams possibly with multi-disciplinary backgrounds as a leader or a member" and "Group project will fulfill the need to assess intended learning outcome d". . Therefore, we prefer that students can complete the project in group so that the course can fulfill the requirements specified in the subject description form. If there are genuine difficulties/justification that you want to complete it individually, please email me for discussion.

References

- <https://www.deeplearning.ai/courses>